Probability Solutions

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October 27, 2026

At a purely formal level, one could call probability theory the study of measure spaces with total measure one.

-Terence Tao

Problems on Probability

1

We win by switching iff the initial box we chose was empty. This has a 2/3 chance, therefore switching wins two thirds of the time. The game is worth 2000/3 dollars by executing this strategy.

2

There are four possibilities for the two cards drawn in order. The probability of each possibility can be computed:

$$QQ = KK = \frac{1}{2} \cdot \frac{3}{7} = \frac{3}{14} \qquad QK = KQ = \frac{1}{2} \cdot \frac{4}{7} = \frac{2}{7}$$

(a) If the dealer tells you atleast one card is a queen then we either have QQ, QK or KQ. The probability of two queens is

$$\frac{3/14}{3/14 + 2/7 + 2/7} = \frac{3}{11}$$

(b) If the dealer tells you that atleast one card is a red queen then the possibilities are a bit different. We still either have QQ, QK, or KQ. However 5/6ths of the ways to draw QQ yield a red queen. Furthermore only 1/2 of the ways to draw QK or KQ have a red queen. Therefore the new probability is

$$\frac{(3/14) \cdot (5/6)}{(3/14) \cdot (5/6) + (2/7) \cdot (1/2) + (2/7) \cdot (1/2)} = \frac{5}{13}$$

(c) If at least one card is the queen of hearts then we have 1/2 possibilities for QQ, and 1/4 for QK and KQ. Therefore the new probability is:

$$\frac{(3/14) \cdot (1/2)}{(3/14) \cdot (1/2) + (2/7) \cdot (1/4) + (2/7) \cdot (1/4)} = \frac{3}{7}$$

3

Let $P(n)$ be the probability that the number of heads is even after throwing n coins. Then the probablity that the number of heads is odd is $1 - P(n)$ and we can derive:

$$P(0) = 1 \qquad P(n+1) = (1-p)P(n) + p(1 - P(n)) = (1-2p)P(n) + p$$

By mindlessly applying this formula a bunch of times recursively we can see that:

$$\begin{aligned} P(n) &= (1-2p)^n P(0) + (1-2p)^{n-1}p + (1-2p)^{n-2}p + \cdots + p \\ &= (1-2p)^n + p\sum_{i=0}^{n-1}(1-2p)^i \\ &= (1-2p)^n + p\frac{1-(1-2p)^n}{1-(1-2p)} \\ &= (1-2p)^n + \frac{1-(1-2p)^n}{2} \\ &= \frac{1+(1-2p)^n}{2} \end{aligned}$$

Which one can check satisfies the reccurance we derived and the initial condition.

4

I would write up this problem, but the question reduces to Q6 with $\alpha = 15$ and $\beta = 60$. Furthermore the argument to prove the result is the exact same. Therefore plug in to obtain:

$$\frac{7}{16}$$

5

Let $E(x)$ be the expected number of turns it takes to exceed the number x. We are asked to compute $E(1)$. We can derive a recursive formula for $E(x)$ by integrating over all possible possibilities of the number that is chosen. In the case that the number is greater than x then the game ends so we do not need to integrate these possibilies.

$$E(x) = \int_0^x E(x-y)\,dy + 1 = \int_0^x E(y)\,dy + 1$$

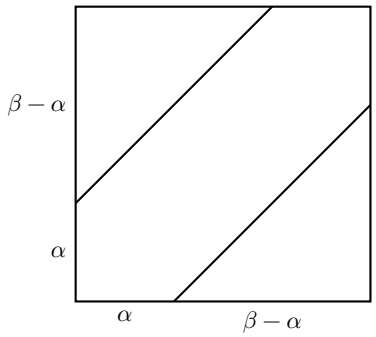
We can solve this by differentiating both sides with respect to x yielding:

$$E'(x) = E(x) \implies E(x) = Ce^x$$

Plugging back in yields $C = 1$ and $E(x) = e^x$. Then plug in 1 to yield $f(1) = e$ as desired.

6

The possibility space looks like a square with side length β:



However we want the regions $y > x + \alpha$ and $y < x - \alpha$. These will be the triangles pictured. The total area of the square is β^2 and the area "cut out" by the triangles is $\beta - \alpha^2$. Therefore the probability of the distance being larger or smaller than α is

$$\frac{(\beta-\alpha)^2}{\beta^2} \qquad \frac{\alpha(2\beta-\alpha)}{\beta^2}$$

respectively.

7

We draw a square and see which points (x, y) satisfy the condition. This condition is equivalent to:

$$\frac{x}{y} \in \left[0, \frac{1}{2}\right] \cup \left[\frac{3}{2}, \frac{5}{2}\right] \cup \left[\frac{7}{2}, \frac{9}{2}\right] \cup \cdots$$

If we visualize each of these regions on the square they will be the triangles cut out by the lines with these slopes. The first region has area $1/4$, and we will consider it separately since the triangle has its base on the right edge. The other triangles have their base on the top edge of the rectangle. The line $y = \alpha x$ will intersect the line $y = 1$ at $x = 1/\alpha$ so the areas sum to:

$$\frac{1}{3} - \frac{1}{5} + \frac{1}{7} - \frac{1}{9} + \cdots$$

This looks like a Taylor series, but we need to figure out which one. The π in the problem statement indicates that it could be an inverse trig function. Using the identity

$$\operatorname{atan} x = \int \frac{1}{1+x^2}\, dx \qquad \frac{1}{1+x^2} = 1 - x^2 + x^4 + \cdots$$

we can derive

$$\operatorname{atan} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

so the solution is $1/4 + (1 - \operatorname{atan}(1))$ or:

$$\frac{5}{4} - \frac{\pi}{4}$$

8

Note that the shape is not convex iff one of the four points lies in the triangle formed by the other 3. Therefore we are essentially asked to compute the expected value of the area of a random triangle inside a circle, and then multiply by 4. Let $E_0(r)$ be the expected area for a circle of radius r. First, let $E_i(r)$ be the expected value of the area with i points chosen on the boundary and $3 - i$ chosen in the interior. We would like to understand how $E_0(r)$ changes when de change r by some dr. There are two types of triangles:

- Those which have a point on the new area dA.
- Those which lie in the original circle.

To first order the propertion of triangles in the first category is $3\frac{dA}{A}$ and the propertion in the second is the complement. The expected area of triangles in the first category is $E_1(r)$ and in the second is $E_0(r)$. Therefore

$$E_0(r + dr) = \frac{3dA}{A} E_1(r) + \left(1 - \frac{3dA}{A}\right) E_0(r)$$

$$= E_0(r) + \frac{3dA}{A}(E_1(r) - E_0(r))$$

$$dE_0 = \frac{3}{A}(E_1(r) - E_0(r))dA$$

which is a case of the *Crofton's differential equation*. Now note by homogeneity

$$\mathbb{E}_i(r) = C_i r^2$$

for some constants. Using the equation to differentiate wrt r we see that

$$2C_0 r = \frac{3}{\pi r^2}(C_1 r^2 - C_0 r^2)2\pi r \implies C_0 = \frac{3}{4}C_1$$

To determine C_1, it suffices to compute $E_1(1)$ for the unit disk. Place the unit circle with centre at $(0,1)$ so that it passes through the origin, and fix the boundary vertex at $P = (0,0)$. Any interior point has polar coordinates (ρ, θ) about P with

$$0 \le \theta \le \pi, \qquad 0 \le \rho \le 2\sin\theta,$$

and the area element is $dA = \rho\, d\rho\, d\theta$. If $Q = (\rho_1, \theta_1)$, $R = (\rho_2, \theta_2)$, then

$$[PQR] = \frac{1}{2}\,\rho_1\rho_2 \big|\sin(\theta_2 - \theta_1)\big|.$$

Since the two interior points are independent and uniform, their joint density is $\frac{1}{\pi^2}\rho_1\rho_2\, d\rho_1 d\theta_1 d\rho_2 d\theta_2$, so

$$E_1(1) = \frac{1}{\pi^2}\int_0^{\pi}\int_0^{\pi}\int_0^{2\sin\theta_1}\int_0^{2\sin\theta_2} \frac{1}{2}\,\rho_1^2\rho_2^2\,\big|\sin(\theta_2 - \theta_1)\big|\, d\rho_1 d\rho_2 d\theta_1 d\theta_2.$$

Using symmetry ($\theta_1 < \theta_2$) and doing the ρ-integrals first,

$$E_1(1) = \frac{64}{9\pi^2}\int_0^{\pi}\int_0^{\theta} \sin^3\theta\, \sin^3\phi\, \sin(\theta - \phi)\, d\phi\, d\theta.$$

Set

$$I = \int_0^{\pi}\int_0^{\theta} \sin^3\theta\, \sin^3\phi\, \sin(\theta - \phi)\, d\phi\, d\theta.$$

Using $\sin(\theta - \phi) = \sin\theta\cos\phi - \cos\theta\sin\phi$,

$$I = \int_0^{\pi} \sin^3\theta \Big(\sin\theta \int_0^{\theta} \sin^3\phi\cos\phi\, d\phi - \cos\theta \int_0^{\theta} \sin^4\phi\, d\phi\Big)\, d\theta.$$

We have

$$\int_0^{\theta} \sin^3\phi\cos\phi\, d\phi = \frac{\sin^4\theta}{4}, \qquad \sin^4\phi = \frac{3}{8} - \frac{1}{2}\cos 2\phi + \frac{1}{8}\cos 4\phi,$$

so

$$\int_0^{\theta} \sin^4\phi\, d\phi = \frac{3}{8}\theta - \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta.$$

Substituting these and integrating termwise over $\theta \in [0, \pi]$ yields

$$I = \frac{35\pi}{256}.$$

Therefore

$$E_1(1) = \frac{64}{9\pi^2} I = \frac{64}{9\pi^2}\cdot\frac{35\pi}{256} = \frac{35}{36\pi}.$$

By homogeneity, $E_1(r) = C_1 r^2$ with $C_1 = \frac{35}{36\pi}$, and hence

$$E_0(r) = \frac{3}{4}E_1(r) = \frac{3}{4}\cdot\frac{35}{36\pi}r^2 = \frac{35}{48\pi}r^2.$$

In particular, for the unit disk,

$$E_0(1) = \frac{35}{48\pi}.$$

and multiplying by 4 and dividing by π gives

$$4 \cdot \frac{35}{48\pi^2} = \frac{35}{12\pi^2}$$

as the probability of the points not forming a convex polygon. Therefore the probability of the points forming a convex polygon is:

$$1 - \frac{35}{12\pi^2}$$

Now I will discuss the general form of Crofton's differential equation. Let f_r be any function on n points in a circle, let E_0 be the expected values of f_r, and E_1 be the expected value with a point on the boundary. As before there are two types of configurations of points when we increase the radius by dr:

- Those which have a point on the new area dA.
- Those which lie in the original circle.

however now the function depends on the radius of the circle. Therefore by computing first order change:

$$\frac{dE_0}{dr} = \frac{n}{A}\left(E_1(r) - E_0(r)\right)\frac{dA}{dr} + \mathbb{E}\left(\frac{\partial f_r}{\partial r}(s)\right)$$

where s is a random configution of points. The first term accounts for the new configurations, which to first order are essentially those which have a point on the boundary. This nasty extra term is why we didn't attempt to use the same method to reduce to E_2 or even E_3.

9

We might consider the numbers modulo 3. There are k zeroes, $k+1$ ones, and k twos. We can essentially place the zeroes anywhere except at the beginning, so we ignore them for a moment.

- If the current sum modulo 3 is 1, the next number must be a 1. The next sum modulo 3 will be 2.
- If the current sum modulo 3 is 2, the next number must be a 2. The next sum modulo 3 will be 1.

Depending on whether we start with a 1 or a 2 we obtain one of the following patterns (of the used numbers):

$$11212121\cdots \qquad 221212\cdots$$

however only the first pattern can use more 1s than 2s. Therefore we are in the first pattern since we have more 1s. The number of ways to assigns the numbers that are 1 is $(k+1)!$ and the number of ways to assign the numbers that are 2 is $k!$. Now we must put in the 0s after the 1s and 2s. There are $2k+1$ "slots" to place k zeroes. By stars and bars the number of ways to do this is

$$\binom{3k}{2k}.$$

The number of ways to assign the numbers that are 0 is $k!$. Therefore the number of good orderings is

$$\binom{3k}{2k}(k+1)!(k!)^2.$$

Since there are $(3k+1)!$ orderings in total, the answer is

$$\frac{\binom{3k}{2k}(k+1)!(k!)^2}{(3k+1)!} = \frac{(k+1)(k!)^2}{(2k)!(3k+1)}.$$

10

We can think of choosing the $x_1, \dots, x_n$ as choosing n random variables $X_1, \dots, X_n$ uniformly from $[0, 1]$ and then ordering them. I define $R(s)$ to be the first chosen point to the right of s, in particular

$$R(s) = x_{i+1} \qquad \text{if } s \text{ is in the inverval } (x_i, x_{i+1}]$$

The Riemann sum is now:

$$\mathbb{E}\left(\int_0^1 f(R(s))\,ds\right) = \int_0^1 \mathbb{E}(f(R(s)))\,ds \tag{1}$$

For $t \geq s$ we define

$$K_s(t) := \text{probability that } R(s) \geq t$$

to be the reversed cumulative distribution function of $R(s)$. This is exactly that every point we choose is in the region $[0, s] \cup [t, 1]$ therefore:

$$K_s(t) = (1 - t + s)^n$$

Taking the negative derivate of this reveals the probability distribution function for $R(s)$:

$$PDF_{R(s)}(t) = -\frac{d}{dt}K_s(t) = n(1 - t + s)^{n-1}$$

Therefore

$$\mathbb{E}(f(R(s))) = \int_s^1 f(t)PDF_{R(s)}(t)\,dt =$$

and plugging into Eq. (1) yields

$$\begin{aligned}
\int_0^1 \int_s^1 f(t)n(1 - t + s)^{n-1}\,dt\,ds &= \int_0^1 \int_0^t f(t)n(1 - t + s)^{n-1}\,ds\,dt \qquad \text{change order of integration} \\
&= \int_0^1 f(t) \int_0^t n(1 - t + s)^{n-1}\,ds\,dt \\
&= \int_0^1 f(t)(1 - (1 - t)^n)dt
\end{aligned}$$

so our polynomial is

$$P(t) = 1 - (1 - t)^n$$

as desired.

11

Let $a_n(t)$ denote the probability that the condition is satisfied given that $x_n = t$. Let $b_n(t)$ denote the probability that the condition is satisfied and $x_n \leq t$. Now note that by checking the last condition:

$$a_n(t) = b_{n-1}(1 - t)$$

however $a_n(t) = b_n'(t)$. Therefore we can differentiate twice:

$$b_n'(t) = b_{n-1}(1 - t) \implies b_n''(t) = -b_{n-2}(t)$$

Now note that $b_0(x) = 1$. Therefore compute manually some terms:

$$b_0(t) = 1 \qquad b_1(t) = t \qquad b_2(t) = t - \frac{t^2}{2}$$

define the generating function

$$H(x, t) = \sum_{n=0}^{\infty} b_n(t)x^n$$

and note that $H(x,1)$ is the desired function. We observe that

$$\frac{\partial^2 H}{\partial t^2} + x^2 H = 0 \implies H(x,t) = A(x)\cos(xt) + B(x)\sin(xt)$$

since this is the equation for a simple harmonic oscillator (wow). When we plug in $t = 0$ all the probabilities go to 0 except b_0. Therefore $A(x) = 1$. Now consider

$$\frac{\partial H}{\partial t} = \sum_{n=0}^{\infty} a_n(t)x^n \implies \frac{\partial H}{\partial t}(x,1) = x$$

since saying that $x_n = 1$ kills all the other probabilities ($x_{n-1} + x_n$ will always be bigger than 1). Plugging this in and solving yields:

$$B(x) = \frac{1+\sin x}{\cos x} \qquad H(x,t) = \cos(xt) + \frac{1+\sin x}{\cos x}\sin(xt)$$

so taking $t = 1$ gives

$$\sum_{n=0}^{\infty} p_n x^n = H(x,1) = \sec(x) + \tan(x)$$

which is an insane result.

12

Suppose the point has coordiantes (x,y) with $x, y \in [-1,1]$. We need to compute the probability that

$$\sqrt{x^2+y^2} \leq \min(1-|x|, 1-|y|).$$

By symmetry we can restrict our analysis to the top right quadrant and assume that $x, y \in [0,1]$. Finally, we will restrict our analysis to below the line $y = x$ so that $y \leq x$. Now write

$$1 - x \geq \sqrt{x^2+y^2} \iff y^2 \leq 1 - 2x$$

which is the condition to be nearer to the center than the edge. We note that this is never satisfied when $x \geq \frac{1}{2}$ so we again restict our analysis to when x is smaller then this value. In this case we have $y \leq \sqrt{1-2x}$, however y can only go up to x. Therefore we have

$$y \leq \min(x, \sqrt{1-2x}) = \begin{cases} x, & x < \sqrt{2}-1 \\ \sqrt{1-2x}, & x > \sqrt{2}-1 \end{cases}$$

Therefore we split this into an integral:

$$2\left(\int_0^{\sqrt{2}-1} x\,dx + \int_{\sqrt{2}-1}^{1/2} \sqrt{1-2x}\,dx\right) = \frac{4\sqrt{2}-5}{3}.$$

13

Write out all the digits of α in base 2, for example:

$$\alpha = 0.100110\ldots$$

and denote α_n to be the nth digit of α after the decimal point. We play a game where on the nth step a player flips a coin, we write out the outcomes

$$\text{outcome of } n\text{th step} = \begin{cases} \text{H} & \begin{cases} \alpha_n = 1 & \text{continue game} \\ \alpha_n = 0 & \text{lose game} \end{cases} \\ \text{T} & \begin{cases} \alpha_n = 1 & \text{win game} \\ \alpha_n = 0 & \text{continue game} \end{cases} \end{cases}$$

The probability of the game continuing for n steps is $1/2^n$ so the game ends in a finite number of steps with probability 1. The chance of winning is α by construction. This is because we are essentially randomly generated an irrational number in $(0,1)$ via its base two representation, then we are checking if it is larger or smaller than α.

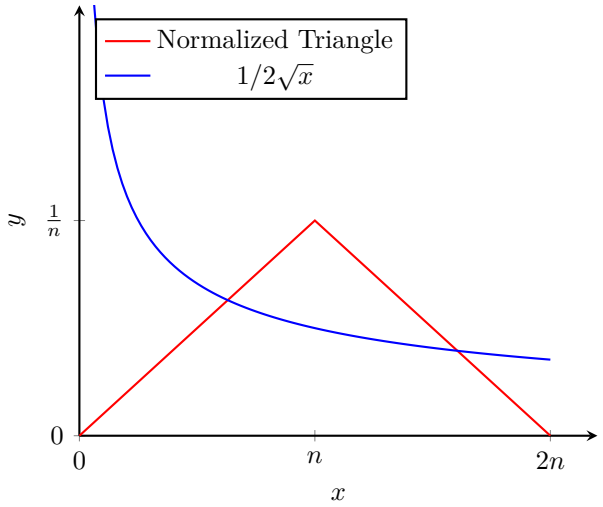
14

Given a point p on the circumference, the allowed region for a point in the center is the rectangle inscribed in the circle with a corner at p. If $p = (\cos\theta, \sin\theta)$ then the rectangle has area $4|\cos(\theta)\sin(\theta)|$ or $2\sin(2\theta)$. The probability that q falls in this rectangle is then $\frac{2\sin(2\theta)}{\pi}$. Using symmetry, we integrate only over the first quarter of the circle to avoid sign errors:

$$\frac{2}{\pi}\int_0^{\pi/2} \frac{2\sin(2\theta)}{\pi}\, d\theta = \frac{4}{\pi}^2$$

15

I'm just going to give a handwaving solution since I don't really like the problem. We are essentially taking the inner product of two density functions.



Where, the red is the PDF of summing two numbers and the blue is the density of the squares. In reality these are discrete, but in the limit it converges to this. The inner product is

$$\begin{aligned}
p_n &\sim \int_0^{2n} T(x)S(x)dx \\
&= \int_0^{n} \left(\frac{x}{n^2}\right)\left(\frac{1}{2\sqrt{x}}\right)dx + \int_n^{2n} \left(\frac{2n-x}{n^2}\right)\left(\frac{1}{2\sqrt{x}}\right)dx \\
&= \frac{1}{2n^2}\int_0^n x^{1/2}dx + \frac{1}{2n^2}\int_n^{2n} \left(2nx^{-1/2} - x^{1/2}\right)dx \\
&= \frac{1}{2n^2}\left[\frac{2}{3}x^{3/2}\right]_0^n + \frac{1}{2n^2}\left[4n\sqrt{x} - \frac{2}{3}x^{3/2}\right]_n^{2n} \\
&= \frac{4(\sqrt{2}-1)}{3\sqrt{n}}
\end{aligned}$$

after a lot of pencil pushing. Therefore

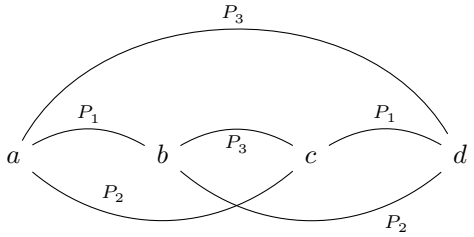
$$\lim_{n\to\infty} p_n\sqrt{n} = \frac{4(\sqrt{2}-1)}{3}.$$

16

The answer is exactly $\frac{2}{3}$, regardless of the number of intervals n (for any $n \geq 2$). We prove this by induction on n. For some notations, I call a pair "universal" if it intersects every other pair, and maximal if it is not strictly contained in another pair. Let p_i denote the pair to i, and let the "spaces" of an interval $[i, j]$ be exactly $|i - j| - 1$ (the number of spaces in between the interval). I claim that when $n > 2$ ignoring the pair $[n, p_n]$ preserves the property of there being a universal interval. There are two things to show:

- Firstly I claim that if $[n, p_n]$ is a universal pair, then there is another universal pair. In the case that $[n, p_n]$ is not maximal, this is clear. If $[n, p_n]$ is maximal then every other pair must have exactly one point inside the interval, so $[n, p_n]$ must have exactly $n-1$ spaces. The only way to achieve this is with the pair $[n, 2n]$ and every other pair $[i, j]$ having $0 \leq i < n$ and $n < j < 2n$. Therefore $[p_{2n-1}, 2n - 1]$ is an interval also intersecting every other interval.
- Now I claim that if there a universal pair among the other intervals, then there is a universal pair. Let $[k, p_k]$ be a maximal pair, universal among the other interals. It must have at least $n - 2$ spaces since every other interval (besides $[n, p_n]$) must have a point inside it. The only way for an interval of this size to "fit" and not contain n is if it is $[n + 1, 2n]$. In this case every other pair $[i, j]$ must have $0 \leq i < n$ and $n + 1 < j < 2n$. Therefore $[p_{2n-1}, 2n - 1]$ intersects every interval including $[n, p_n]$.

Now, consider the base case of four points a, b, c, d: Because the first $n - 2$ pairings did not restrict the pairing of the final four points, these four points are paired uniformly at random in one of three possible ways. We can visualize these three pairings (P_1, P_2, P_3) using the following layout:



We can see that 2 out of the three matching satisfy the universal interval condition. Therefore the answer is always 2/3.

17

Let X and Y be the temperatures in Chicago and Detroit, respectively. We are given $P(X = 70)$, $P(Y = 70)$, and $P(\max(X, Y) = 70)$, and we want to find $P(\min(X, Y) = 70)$.

Since we are only concerned with how the temperatures compare to 70°, we can logically partition the sample space into a 3×3 grid based on whether X and Y are less than, equal to, or greater than 70. Let $p_{i,j}$ denote the probability of each joint state:

	$X < 70$	$X = 70$	$X > 70$
$Y > 70$	$p_{<,>}$	$p_{=,>}$	$p_{>,>}$
$Y = 70$	$p_{<,=}$	$p_{=,=}$	$p_{>,=}$
$Y < 70$	$p_{<,<}$	$p_{=,<}$	$p_{>,<}$

From this grid, we can express the marginal probabilities as sums of these disjoint regions:

- $P(X = 70)$ is the sum of the middle column: $p_{=,>} + p_{=,=} + p_{=,<}$
- $P(Y = 70)$ is the sum of the middle row: $p_{<,=} + p_{=,=} + p_{>,=}$

The event $\max(X, Y) = 70$ occurs if the highest of the two temperatures is exactly 70. This means one temperature is exactly 70 and the other is 70 or lower. This corresponds to the bottom-left "L" shape of our middle cross:

$$P(\max(X, Y) = 70) = p_{<,=} + p_{=,=} + p_{=,<}$$

The event $\min(X, Y) = 70$ occurs if the lowest of the two temperatures is exactly 70. This means one temperature is exactly 70 and the other is 70 or higher. This corresponds to the top-right inverted "L" shape of our middle cross:

$$P(\min(X, Y) = 70) = p_{=,>} + p_{=,=} + p_{>,=}$$

Notice what happens when we add the probabilities of the middle column and the middle row. The center cell $p_{=,=}$ is naturally counted twice:

$$\begin{aligned} P(X = 70) + P(Y = 70) &= (p_{=,>} + p_{=,=} + p_{=,<}) + (p_{<,=} + p_{=,=} + p_{>,=}) \\ &= (p_{<,=} + p_{=,=} + p_{=,<}) + (p_{=,>} + p_{=,=} + p_{>,=}) \\ &= P(\max(X, Y) = 70) + P(\min(X, Y) = 70) \end{aligned}$$

By rearranging this additive relationship, we can quickly solve for our target probability:

$$P(\min(X, Y) = 70) = P(X = 70) + P(Y = 70) - P(\max(X, Y) = 70)$$

18

Suppose that I arrange 37 slots around a circle and randomly select 7 of them. To the first slot I selected, I write the number 0, and then proceed move counterclockwise around the circle writing the numbers $1, 2, \ldots, 36$. From this procedure two things are clear

- The maximum number written in a slot I chose is equivalent to the MEGABUCKS lottery.
- The gap between two adjascent selected slot is the same, and has expected value $37/7$.

The slot corresponding to the maximum number I chose is adjascent to 0, with expected gap $37/7$. Therefore the answer is

$$37 - 37/7 = \frac{6}{7} \cdot 37$$

19

Construct the Markov chain with nodes $X_0, X_1, X_2 \ldots$ and transition probabilities:

- Node X_0 has transition probability a_i to jump to node X_{i-1}.
- Node X_i for $i \neq 0$ has probability 1 to jump to node X_{i-1}.

At step 0 we start on X_0. The interpretation is the following. Let $\{S_i\}$ be the set of partial sums in the rolling process. We define

$$Y_n = \min\{S_i - n \mid S_i - n \geq 0\}$$

to be the overshoot past n. Then at the nth step of the Markov chain, being at the node X_i means we overshot n by i. In particular:

$$p_n = \text{probability of being at } X_0 \text{ after } n \text{ steps}$$

Since the Markov chain is irreducibe and aperiodic, the limit $\lim_{n\to\infty} p_n$ exists. Moreover the limit is exactly the recipricol of the expected amount of steps to return to X_0. This is exactly

$$\left(\sum i a_i\right)^{-1}$$

which is what we would expect.

20

Define

$$A_n, B_n, C_n$$

to be the *unarranged* row sums along the first n columns, minus $2n$. We have that

$$\begin{bmatrix} A_{n+1} \\ B_{n+1} \\ C_{n+1} \end{bmatrix} = \begin{bmatrix} A_n \\ B_n \\ C_n \end{bmatrix} + \text{some permutation of } -1, 0, 1$$

Since the permutations of $-1, 0, 1$ span a 2 dimensional space (the sum of coordinates is fixed), it suffices to look at only the first two coordinates. The possible steps for $\begin{bmatrix} A \\ B \end{bmatrix}$ are:

$$\pm \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \pm \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad \pm \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

These are also exactly the set of vectors corresponding to the problem condition, therefore we need to show that

$$P\left(\pm \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \pm \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \pm \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \geq 4P(0)$$

however by permutation symmetry, we have an equal chance of landing on any of these vectors, therefore what we are really trying to show is

$$6P\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \geq 4P(0)$$

This is a result of the following theorem:

Theorem 1.1. Local Central Limit Theorem for Lattice Walks Let $X_1, X_2, ...$ be a sequence of independent, identically distributed random vectors taking values in the d-dimensional integer lattice $\mathbb{Z}^d$. Let $S_n = \sum_{i=1}^n X_i$.
Assume the following conditions are satisfied:

(a) **Finite Second Moments:** $\mathbb{E}[\|X_1\|^2] < \infty$ and $\Sigma = \mathrm{Cov}(X_1)$ is strictly positive-definite.

(b) **Strong Aperiodicity:** The characteristic function $\varphi(\theta) = \mathbb{E}[\exp(i\theta^\mathsf{T} X_1)]$ satisfies $|\varphi(\theta)| = 1$ if and only if $\theta \in 2\pi\mathbb{Z}^d$.

Then, as $n \to \infty$, the probability distribution of S_n satisfies:

$$\sup_{x \in \mathbb{Z}^d} \left| \mathbb{P}(S_n = x) - \frac{1}{(2\pi n)^{d/2}\sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2n}(x - n\mu)^\mathsf{T} \Sigma^{-1} (x - n\mu) \right) \right| = o(n^{-d/2})$$

where $\mu = \mathbb{E}[X_1]$.

In particular

$$p_n(v) \sim \frac{\sqrt{3}}{2\pi n} \qquad \text{as } n \to \infty.$$

for any reachable vector. Therefore in the limit the ratio is 6 to 1. I would recommend checking out the proof of the claim, it's some cool Fourier stuff.

21

Let $P(k)$ be the probability that $\sum_{i=1}^k (x_i^2 + y_i^2)$ is less than one. By linearity of expecation what we are trying to compute is

$$\sum_{k=0}^{\infty} P(k)$$

However, $P(k)$ expresses that a random point we choose in $[0,1]^{2k}$ lies in the unit ball of $\mathbb{R}^{2k}$ By symmetry we might as well pick the point from $[-1,1]^{2k}$ so that

$$P(k) = \frac{\text{Vol}(2k \text{ unit ball})}{\text{Vol}(2k\text{-cube of side length 2})} = \frac{\pi^k}{4^k k!}$$

by the relevant formulas. Then the infinite series we are trying to evaluate becomes exactly the power series of e^x evaluated at $\pi/4$ and the answer is $e^{\pi/4}$.

22

Let E be the expected number of flips to get one more head than tail.

- On the first flip, we have a $1/2$ chance of getting a heads and ending the game. This contributes $\frac{1}{2}$ to the expected value.
- Otherwise, we need to get one more heads then tails in order to get back where we started. This contributes $\frac{1}{2}(E+E+1)$.

Therefore

$$E = E + 1$$

and the expected value of the game diverges to infinity. This is interesting since we can be very confident that the game ends eventually.

Problem on Probability Games

23

Repeat the following:

- Pick the first feasible number available to you.
- Pick the second feasible number available to you.

We lose exactly if the correct number is 0 modulo 3 (picking it on the second step). This is better than $2/3$ odds of winning as desired.

24

There is no such strategy. To see this let $f(a,b)$ be the expected profit when there are a winning cards and b losing cards. I claim that:

$$f(a,b) = \frac{a}{a+b}$$

It is clear this is the probability we obtain if we say stop instantly. Now I proceed by induction on $a+b$. If we do not stay stop our expected value is

$$\frac{a}{a+b} f(a-1,b) + \frac{b}{a+b} f(a,b-1) = \frac{a}{a+b}\frac{a-1}{a+b-1} + \frac{b}{a+b}\frac{a}{a+b-1} = \frac{a}{a+b}$$

so our choice didn't matter. By setting $a = b$ we see the probability of winning is always a half.

The answer is also seen as an immediate consequence of some knowledge on Martingales. In particular, the expected proportion of red cards after a draw is equal to the current propertion. Therefore it is a Martingale whose stopping time is clearly bounded. The goal is to have a stopping strategy that maximizes the expected value of this Martingale at the stop time. However by the optional stopping theorem, your choice of when to stop doesn't matter.

25

Let $M_{a,b}$ be the maximum amout of money that you are guarenteed to win with a red cards and b blue cards. On the draw if you stake x on red (negative values correspond to picking blue) and then play perfectly your guarenteed winning is either

$$(1+x)M_{a-1,b} \qquad (1-x)M_{a,b-1}$$

and $M_{a,b}$ is the minimum of these two values for the choice of x that maximizes it. In order to maximize the minimum of two linear equations we just set them equal:

$$(1+x)M_{a-1,b} = (1-x)M_{a,b-1} \implies x = \frac{M_{a-1,b} - M_{a,b-1}}{M_{a-1,b} + M_{a,b-1}} \implies M_{a,b} = \frac{2M_{a-1,b}M_{a,b-1}}{M_{a-1,b} + M_{a,b-1}} \tag{1}$$

Note that this is exactly the Harmonic mean of $M_{a-1,b}$ and $M_{a,b-1}$. The motives us to define

$$R_{a,b} := \frac{2^{a+b}}{M_{a,b}} \implies R_{a,b} = R_{a-1,b} + R_{a,b-1}$$

which is a much better form of Eq. (1). We note that $M_{k,0} = M_{0,k} = 2^k$ which shows that $R_{k,0} = R_{0,k} = 1$. Therefore the $R_{i,j}$ satisfy the exact same initial conditions and recurrence as the binomial coefficients so:

$$R_{a,b} = \binom{a+b}{a} \implies M_{a,b} = 2^{a+b}\binom{a+b}{a}^{-1}$$

We can plug in $a = b = 26$ to solve the problem. Numerically

$$M_{26,26} \approx \$9.08133$$

which is better than \$2.

26

I assume that Alice does not want Bob to win, otherwise it is very easy for them to coordinate a strategy. In this case, Bob can always get at least a bit better than 50%.

If Bob knows Alice's probability distribution, he may pick a threshold number T which lies between some pair Alice might pick. Then he swaps if the picked number is less than T. This esures that Bob picks correctly on those pairs, and on other pairs he still has a 50% win rate.

In the case that Bob does not know Alice's distribution, he picks T from a distribution with positive density everywhere. Making sure Alice cannot be guarenteed to avoid his slight advantage.

27

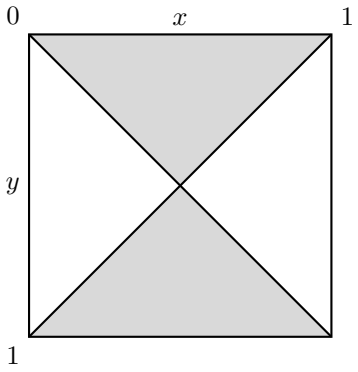
Bob in fact cannot do better than to break even. Suppose that Alice executes the following strategy:

- If the sum of the numbers is less than 1, show the bigger number.
- Otherwise show the smaller number.

Now let's suppose Bob receives the number x.

- If x is bigger, the other number is in the interval $[0, 1-x] \cap [0, x]$ which has length $\min(x, 1-x)$.
- If x is smaller, the other number is in the interval $[1-x, 1] \cap [x, 1]$ which has length $\min(x, 1-x)$.

therefore the probability of each possibility is the same. This corresponds to the following diagram:



where the dark area corresponds to choosing the number x to show Bob. The intersection of any vertical line with the dark area has the same length in the $x > y$ and $y < x$ region. Similarily the intersection of any horizantal line with the light area has the same length in the $x > y$ and $x < y$ regions.

Now we must show that Bob can in fact break even. This is by ignoring the number and randomly deciding when to switch.

More Problems

28

This is a linearity of expectation problem. Given each value from 1 to 2013, the expected probability of choosing it at least once is

$$1 - \left(1 - \frac{1}{2013}\right)^{2013}$$

and therefore to get the total expected value we multiply by the number of distinct elements.

$$2013 - 2013\left(\frac{2012}{2013}\right)^{2013} \approx 2013(1 - e^{-1})$$

29

Define

$$p_n^k := \text{probability that } M \equiv n \pmod{k}$$

Since 100 is divisible by 2, we see that each of $\{a, b, c, d, e, f\}$ has equal chance to be even or odd. From the $32f$ term we see that

$$a \equiv b \pmod{32} \implies p_a^{64} = p_b^{64}$$

so it suffices to compute p_n^{32}. Moreover the $16e$ term further means

$$a \equiv b \pmod{16} \implies p_a^{32} = p_b^{32}$$

and it suffices to compute p_n^{16}. Making arguments of the form, we eventually see that the probability only depends on p_n^1, so every residue has the same probability. Therefore the expected value is

$$\frac{1}{64}\sum_{k=0}^{63} k = \frac{1}{64}\frac{63 \cdot 64}{2} = \frac{63}{2}$$

by the triangular numbers formula.

30

Yes. We model the numbers on the blackboard as a Markov chain $(X_n)_{n\geq 0}$ with $X_0 = 2019$ and transition probabilities:

$$\mathbb{P}(X_{n+1} = k \mid X_n = a) = \frac{1}{\lceil 2.01a \rceil + 1} \quad \text{for } k \in \{0, 1, \ldots, \lceil 2.01a \rceil\}$$

Note that 0 is an absorbing state; if $a = 0$, the next number is chosen uniformly from $\{0\}$, meaning $X_{n+1} = 0$ with probability 1. To show that the process reaches 0 in finite expected time, we use a potential function and Foster's Theorem. Let our potential function be the harmonic sequence, $f(x) = H_x = \sum_{j=1}^{x} \frac{1}{j}$ for $x \geq 1$, and $f(0) = 0$. Let $m = \lceil 2.01a \rceil$. Given $X_n = a$, the expected value of the potential function at the next step is:

$$\mathbb{E}[f(X_{n+1}) \mid X_n = a] = \frac{1}{m+1} \sum_{k=0}^{m} H_k$$

Using the identity $\sum_{k=0}^{m} H_k = (m+1)H_m - m$, we substitute to find:

$$\mathbb{E}[f(X_{n+1}) \mid X_n = a] = \frac{(m+1)H_m - m}{m+1} = H_m - \frac{m}{m+1}$$

The drift of the process, $\Delta(a)$, is defined as the expected change in the potential function:

$$\Delta(a) = \mathbb{E}[f(X_{n+1}) \mid X_n = a] - f(a) = H_m - H_a - \frac{m}{m+1}$$

For large values of x, the harmonic numbers can be approximated by $H_x \approx \ln(x) + \gamma$, where γ is the Euler-Mascheroni constant. Therefore, as $a \to \infty$:

$$H_m - H_a \approx \ln(m) - \ln(a) = \ln\left(\frac{m}{a}\right)$$

Since $m = \lceil 2.01a \rceil$, the ratio $\frac{m}{a} \to 2.01$. Additionally, $\frac{m}{m+1} \to 1$. Thus, the asymptotic drift is:

$$\lim_{a \to \infty} \Delta(a) = \ln(2.01) - 1$$

Because $2.01 < e \approx 2.718$, it is strictly true that $\ln(2.01) < 1$, which means $\ln(2.01) - 1 < 0$.

Let $\epsilon > 0$ such that $\ln(2.01) - 1 = -2\epsilon$. There exists some threshold integer A such that for all $a \geq A$, the drift satisfies:

$$\Delta(a) \leq -\epsilon < 0$$

By Foster's Theorem, because we have a non-negative potential function with a negative drift strictly bounded away from zero outside a finite set of states, the expected time to reach that finite set (and ultimately the absorbing state 0) is finite. Let T_0 be the number of steps until absorption. We have proven that:

$$\mathbb{E}[T_0 \mid X_0 = 2019] < \infty$$

Because the expected time to absorption is finite, the cumulative probability of absorption approaches 1 as the number of steps N approaches infinity:

$$\lim_{N \to \infty} \mathbb{P}(T_0 \leq N) = 1$$

Therefore, by the definition of a limit, there must exist some finite integer N for which $\mathbb{P}(T_0 \leq N) \geq 0.99$.

31

Suppose for the sake of contradiction the $\mathcal{A}$ is a σ-algebra with countably infinite cardinality. For each $x \in X$ define its atom

$$C_x := \bigcap_{\substack{A \in \mathcal{A} \\ x \in A}} A$$

which is a measurable set because it is a countable intersection. I claim that if $C_x \neq C_y$ then they are disjoint. Note that $C_x = C_y$ if and only if there exists no measurable set which separates them. On the other hand if there exists some $B \in \mathcal{A}$ which separates x and y (by taking complement we assume $x \in B$ and $y \notin B$)

$$C_x \subseteq B \qquad C_y \subseteq B^c$$

proving disjointness. In particular the atoms partition the topological space. Now we have two cases:

- If there are only finitely many distinct atoms, then since any measurable set is the union of the atoms it contains, the σ-algebra must be finite, a contradiction.
- If there are infinitely many atoms, then any countable union of them is a measurable set. Since they are disjoint, there are at least $2^{\mathbb{N}}$ distinct measurable sets built in this way, a contradiction.

We also get for free that the possible values on the cardinality are 2^m where m is the number of atoms.

32

Suppose that Kevin randomly arranges all the cookies in an ordered line. On each day, he looks at the cookie that is first in line, then he eats that one and all the cookies which are labeled with a subset of that cookie. From the procedure we observe

- Since the initial arrangement was random, this is equivalent to randomly selecting a cookie every day
- The number of days that pass is the number of cookies he looks at.
- A cookie is looked at exactly when no cookie earlier in the line is labeled with a superset of that cookie.

Therefore to compute the expected number of days, we compute the probability that each cookie is not of subset of an earlier cookie, and then sum over all the cookies.

Suppose a cookie is labeled with a subset of size k, then there are 2^{n-k} cookies (including itself) that are labeled with a superset of this. The probability that this cookie appears first out of all these cookies is exactly $1/2^{n-k}$. Therefore compute

$$\begin{aligned}
\mathbb{E}(\text{days}) &= \sum_{C \in \text{cookies}} P(C \text{ is looked at}) \\
&= \sum_{k=1}^{n} \sum_{C \in \text{cookies with size } k \text{ subset}} 1/2^{n-k} \\
&= \sum_{k=1}^{n} \binom{n}{k} 2^{k-n} && \text{there are } \binom{n}{k} \text{ subsets of this size} \\
&= 2^{-n}(1+2)^n - 2^{-n} && \text{binomial theorem, excluding 0 term} \\
&= (3/2)^n - 2^{-n}
\end{aligned}$$